

response times, including but not limited to connected vehicles, smart cities, and industrial automation.

Decentralised model training is a collaborative learning approach where the training process is distributed across multiple edge devices or nodes instead of relying on a centralised server. In traditional centralised training, all the raw data is sent to a central server, where the model is trained. However, in a decentralised model, training occurs locally on individual devices, and only model updates are shared.

The rise of edge computing and the advent of 5G networks have made decentralised model training more feasible and efficient. Edge devices, equipped with sufficient computational power, can now participate in training models collectively.

Key components of decentralised model training

Local training: Each edge device performs local model training using its own dataset. This dataset may contain unique patterns and characteristics specific to that device's environment or user behaviour.

Model update sharing: After local training, only the model updates, rather than the raw data, are shared among the devices. This ensures data privacy and security, as sensitive information remains on the local device.

Aggregation and updating: The central server, or a coordinating entity, aggregates the model updates received from various devices. The global model is then updated based on the aggregated information, reflecting the collective knowledge learned from the decentralised network.

Federated learning

Decentralised model training is closely related to federated learning, a subset of this approach. In federated learning, the model is trained across decentralised edge devices, and the global model is

iteratively improved without exchanging raw data. This iterative process allows for continuous learning and adaptation to local conditions.

To elaborate further, federated learning extends the notion of decentralised model training by permitting edge devices to educate a global model using the insights they gain from their local datasets. Following this, updates are made to the global model without requiring the transmission of raw data to a central server. This methodology not only guarantees confidentiality but also promotes ongoing education and adjustment to regional circumstances.

Advantages of decentralised model training

Privacy-preserving: One of the primary advantages of the decentralised model training is its privacy-preserving nature. Since raw data remains on local devices, there is no need to expose sensitive information to a central server, mitigating privacy concerns associated with centralised approaches.

Reduced latency: The localised training and model updating processes significantly reduce latency. This is crucial in applications where real-time decision-making is essential, such as autonomous vehicles, healthcare monitoring, and augmented reality.

Resource efficiency: Decentralised model training optimises resource usage by distributing the computational load across edge devices. This is particularly beneficial for devices with limited processing power, as it prevents overloading and ensures efficient utilisation of available resources.

Adaptability to dynamic environments: The decentralised nature of model training allows the system to adapt to dynamic and diverse environments. Localised learning captures nuances and changes specific to each device, leading to more robust and adaptive models.

Challenges and considerations in decentralised model training

Communication overhead: While decentralised model training reduces data transfer, there is still a need for communication between devices and the central server. Efficient communication protocols must be implemented to minimise overhead.

Heterogeneous data: Devices in a decentralised network may have heterogeneous data distributions. Ensuring that the global model captures representative information from all devices requires careful aggregation strategies.

Security concerns: While decentralised model training enhances privacy, it introduces new security challenges. Secure communication channels and robust encryption mechanisms are essential to protect against potential threats and attacks.

Ensuring privacy in federated learning settings

Balancing privacy with utility in this decentralised setting requires careful navigation. Threats to privacy in federated learning are:

- **Inference attacks:** Even without raw data, patterns in aggregated updates or model parameters can leak information. Imagine a federated learning system for spam filtering. An attacker could infer which devices receive specific types of emails, revealing sensitive topics like health or finances.
- **Model inversion:** Trained models might hold the key to reconstructing individual data points. An attacker could reverse engineer a credit fraud detection model to uncover financial transactions of specific users, posing a significant privacy risk.
- **Differential privacy trade-off:** To protect individuals, noise is added to data. However, this can decrease model accuracy. Think of a personalised shopping recommendation system – too much